

COMPREHENSIVE CHAPTER WORKSHEET: PRINCIPLES OF INHERITANCE AND VARIATION

PU-II Biology (Class 12) • Complete Chapter Mastery Assessment

Student Name: _____

Roll / Reg. No: _____

Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

- Which of the following represents a pair of dominant traits in garden pea (*Pisum sativum*)?
(A) Violet flower and Yellow pod
(B) Violet flower and Inflated pod
(C) White flower and Round seed
(D) Terminal flower and Green pod
- In snapdragon (*Antirrhinum majus*), crossing pure red (RR) with pure white (rr) yields pink (Rr) F₁ due to:
(A) Co-dominance
(B) Incomplete dominance
(C) Pleiotropy
(D) Polygenic inheritance
- Which organism exhibits female heterogamety (ZZ male and ZW female)?
(A) *Drosophila melanogaster*
(B) Birds / Domestic fowl
(C) Grasshopper
(D) Honeybee
- A male patient with karyotype 47, XXY exhibiting tall stature and gynaecomastia is diagnosed with:
(A) Down's syndrome
(B) Turner's syndrome
(C) Klinefelter's syndrome
(D) Cri-du-chat syndrome

SECTION B: Fill in the Blanks (1 Mark Each)

- The graphical representation developed to calculate the probability of genotypes in genetic crosses is
- Human ABO blood groups are controlled by gene *I* which has alleles, producing 6 genotypes and 4 phenotypes.
- In honeybees, haploid male drones develop from unfertilized eggs through the process of
- Sickle cell anaemia is caused by substitution of Glutamic acid by at the 6th position of the beta-globin chain.

SECTION C: Very Short Answer Questions (2 Marks Each)

9. [2M] What is a test cross? State its significance in genetics.

10. [2M] Differentiate between Male Heterogamety and Female Heterogamety with examples.

11. [2M] Define 'Linkage' and 'Recombination' as described by T.H. Morgan.

SECTION D: Short Answer Questions (3 Marks Each)

12. [3M] State the three key principles of Sutton-Boveri Chromosomal Theory of Inheritance.

13. [3M] A child has blood group 'O'. The father is blood group 'A' and mother is blood group 'B'. Work out the parental genotypes and possible blood groups of future offspring.

14. [3M] Explain the biochemical basis and clinical symptoms of Phenylketonuria (PKU).

SECTION E: Long Answer / Essay Questions (5 Marks Each)

15. [5M] Describe Mendel's Dihybrid cross in garden pea with seed shape and seed colour. Give the Punnett square, phenotypic ratio (9:3:3:1), and state the Law of Independent Assortment.

16. [5M] Compare Down's syndrome, Turner's syndrome, and Klinefelter's syndrome with respect to: (a) Chromosomal abnormality / Karyotype (b) Total chromosome number (c) Key clinical symptoms.